Skin Cancer Classification: Multi-class Diagnosis of 7 Pigmented Skin Lesions

Skin Cancer Classification

ARTIFICIAL INTELLIGENCE CAPSTONE PROJECT PREPARATION

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PROBLEM STATEMENT

Skin cancer is a global health concern, and Melanoma, the deadliest form, requires early and accurate detection for successful treatment. Current clinical diagnosis relies heavily on visual inspection using dermoscopy, which can be subjective and time-consuming. The challenge is compounded by the visual similarity between benign and malignant lesions, and the high variability in the appearance of different lesion types.

This project addresses the need for an accurate, automated, and multi-class classification system capable of differentiating between 7 distinct types of skin lesions from dermoscopic images. Such a system would serve as a crucial computer-aided diagnostic (CAD) tool, reducing diagnostic error rates, increasing efficiency, and improving patient outcomes through earlier intervention.

PROJECT OBJECTIVES

The primary goal of this project is to develop a robust Deep Learning model capable of classifying dermoscopic images into 7 distinct diagnostic categories. The specific objectives are:

1. Multi-Class Classification: Design and train a high-performance Convolutional Neural Network (CNN) model to accurately classify the 7 skin lesion types.
2. Address Data Imbalance: Implement advanced deep learning techniques, such as specialized loss functions and augmentation, to effectively mitigate the severe class imbalance present in the dataset.
3. Achieve Benchmark Performance: Validate the model's performance using professional metrics, aiming for a minimum Macro-Averaged F1-Score of a competitive benchmark (e.g., 0.80) to ensure reliable performance across all 7 classes, including the rare malignant types.

STAKEHOLDERS AND AUDIENCE

Category	Description
Primary Stakeholders	Dermatologists and Clinicians: Will use the resulting model as a decision-support tool for initial screening and diagnosis, potentially improving accuracy and saving time.
Secondary Stakeholders	Medical AI Researchers: The project contributes a robust, well-documented solution to the challenging multi-class skin lesion diagnosis problem.
Target Audience	Computer Vision/Deep Learning Developers: Will be interested in the implementation details, particularly the strategies used for handling imbalanced medical data (e.g., Focal Loss and Transfer Learning).

AI METHODOLOGIES

This project will employ Deep Learning (DL), leveraging the powerful image feature extraction capabilities of Convolutional Neural Networks (CNNs).

A. Core Architecture: Transfer Learning

A Transfer Learning approach will be used by fine-tuning a state-of-the-art CNN architecture (e.g., EfficientNet or DenseNet), which has been pre-trained on a large general image dataset (ImageNet). This strategy accelerates training and ensures robust feature extraction from the dermoscopic images.

B. Advanced Techniques for Imbalanced Data (Complexity Requirement)

To address the known severe class imbalance in the HAM10000 dataset, the following techniques will be implemented:

1. Focal Loss: This specialized loss function will be used to dynamically down-weight the contribution of easy, well-classified examples (the common benign lesions), thereby focusing the model's learning on the hard-to-classify and crucial minority class samples (like Melanoma).
2. Aggressive Data Augmentation: Extensive augmentation (e.g., rotation, flipping, color jitter) will be applied, with a higher probability or intensity targeting the images belonging to the minority classes to artificially balance the training set.

C. Preprocessing

Standard preprocessing will include image resizing and pixel normalization. Crucially, methods for Hair and Artifact Removal will be investigated and applied to clean the dermoscopic images, as artifacts can significantly interfere with feature learning.

D. Model Interpretability

Grad-CAM (Gradient-weighted Class Activation Mapping) heatmaps will be generated to visualize the exact regions of the skin lesion that most influenced the model’s prediction. These heatmaps will be overlaid on the original dermoscopic images and included in the notebook, final report, and video presentation to demonstrate clinical relevance and build trust in the AI system.

In medical AI, a model's high accuracy is often not enough. Clinicians need to understand why the model made a specific prediction (e.g., classifying a lesion as Melanoma).

- Increases Trust: By visually showing that the model is focusing on the lesion area and not background noise or artifacts, you build confidence in the system.
- Debugs the Model: If a heatmap shows the model is looking at a ruler or a hair instead of the lesion itself, you immediately know there's a problem with your preprocessing or training.

DATASET JUSTIFICATION

Criteria	Dataset Used: HAM10000 (Human Against Machine with 10000 training images)	How the Criterion is Met
Public Availability	The HAM10000 dataset is a publicly released, multi-source collection of dermoscopic images.	Freely Accessible. It is the primary benchmark dataset from the ISIC 2018 challenge.
Relevance	It is a representative collection of all important diagnostic categories in pigmented lesions, including malignant and benign types.	Direct Real-World Application. Directly supports the medical objective of skin cancer diagnosis.
Size	The dataset contains 10,015 dermoscopic images across 7 classes. (Initial download size is approx. 6GB).	Sufficient Size. It provides enough data for training and robust evaluation of modern deep CNN architectures.
Complexity	The dataset exhibits a severe class imbalance (e.g., Melanocytic Nevi has over 6,700 images while Dermatofibroma has only 115).	Challenging Dataset. This imbalance *necessitates* the application of advanced AI techniques (Focal Loss, targeted augmentation) to prevent model bias, satisfying the complexity requirement.

The 7 classes for classification are: Actinic Keratoses/Intraepithelial Carcinoma (AKIEC), Basal Cell Carcinoma (BCC), Benign Keratosis-like lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Melanocytic Nevus (NV), and Vascular Lesions (VASC).

Reference: Tschandl, P., et al. (2018). "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions". https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/DBW86T

FEASIBILITY ANALYSIS

The project is highly feasible within the capstone timeline due to the following factors:

- Computational Resources: The use of Google Colaboratory provides free access to necessary computational resources, specifically GPUs, which are required for training deep CNN models on a dataset of this size (6GB).
- Tooling and Libraries: The entire project can be implemented using robust, open-source Python libraries (TensorFlow/Keras or PyTorch for deep learning; Scikit-learn for metrics and data splitting; OpenCV for image preprocessing).
- Risk Mitigation: The primary technical risk is the model overfitting to the majority class (NV), which would yield a high overall accuracy but poor performance on critical malignant classes. This risk is directly mitigated by the planned implementation of Focal Loss and Stratified Data Augmentation, which forces the model to learn from the rare examples.

EXPECTED OUTCOMES AND METRICS

The deliverables and evaluation metrics are chosen to demonstrate successful completion of the project objectives and provide a professional, defensible evaluation of the model's clinical utility.

Expected Outcomes (Deliverables)

1. A fully documented GitHub repository including a clean, runnable Jupyter Notebook (Skin_Cancer_HAM10000_Final.ipynb) with all code and results.
2. A Professional Report (8-10 pages) detailing the methodology, data analysis, and findings.
3. A Video Presentation (10-15 minutes) demonstrating the final prototype.

Evaluation Metrics

Given the imbalanced nature of the dataset, the project will prioritize metrics that account for all classes, not just overall accuracy:

- Primary Metric: Macro-Averaged F1-Score (The most critical metric, as it treats the F1-Score of all 7 classes equally, ensuring reliable diagnosis for both common and rare lesions).
- Secondary Metrics:
    - Accuracy (Overall)
    - Area Under the Curve (AUC) per class.
    - Confusion Matrix (Visualizing misclassifications, especially between benign and malignant lesions).